ASSIGNMENT 6 — QUESTION 9

Q9 - Debugging the Event Loop

JavaScript (ES6+) solution with comments

```
// Q9: Predicted output order BEFORE execution:
// 1. Script start
// 2. Script end
// 3. Promise callback
// 4. Timeout callback

console.log("Script start");

setTimeout(() => {
    console.log("Timeout callback");
}, 0);

Promise.resolve().then(() => {
    console.log("Promise callback");
});

console.log("Script end");

/*
Compare the console output with the prediction.
Explanation:
- "Script start" and "Script end" are synchronous and run first.
- Promise.then() schedules a microtask.
- setTimeout() schedules a task/macrotask, even with a 0 ms delay.
- After synchronous code finishes, the event loop processes pending
  microtasks before moving to the timer task.
*/
```